Linear Algebra
Quiz 3

Day: M T W T F S Date: __/__/20 Day: M T W T F S Date: __/__/20

Muhammad Hassan Khalid 21L-5692

Let
$u_1 (3,1,-1,3)$, $u_2 (1,1,-2,8)$
$u_3 (-5,1,5,-7)$

a $v_1 = u_1 = (3,1,-1,3)$

$v_2 = u_2 = \frac{\langle u_2 - v_1 \rangle v_1}{\|v_1\|^2}$

$v_2 = (1,1,-2,8) - \frac{30}{20}(3,1,-1,3)$

$= (1,1,-2,8) - \left(\frac{3}{3}, \frac{3,3}{2}, \frac{-3}{2}, \frac{4}{2}\right)$

$= \left(-\frac{7}{2}, -\frac{1}{2}, -\frac{7}{2}, \frac{7}{2}\right)$

$v_3 = u_3 = \frac{\langle u_3 \cdot v_1 \rangle v_1}{\|v_1\|^2} = \frac{\langle u_3 v_2 \rangle v_2}{\|v_2\|^2}$

$v_3 = (-5,1,5,7) = \frac{40}{20}(3,1,-1,3) + \frac{25}{27}\left(-\frac{7}{2}, -\frac{1}{2}, -\frac{7}{2}\right)$

$v_3 = (-5,1,5,7) - (6,2,-2,6) + \left(\frac{-175}{74}, \frac{-25}{74}, \frac{-175}{74}, \frac{175}{74}\right)$

$= \left(-5-6-\frac{175}{74}, 1-6-\frac{25}{74}, 5+2-\frac{175}{74}, 7-6+\frac{175}{74}\right)$

$= \left(\frac{84-175}{74}, \frac{-468}{74}, \frac{343}{74}, \frac{249}{74}\right)$

$v_3 = \left(\frac{639}{74}, \frac{-468}{74}, \frac{343}{74}, \frac{249}{74}\right)$

b Orthogonal basis

$\|v_1\| = \sqrt{20}$ $\|v_2\| \sqrt{37}$ $\|3\| \sqrt{}$

$v_1' \left(\frac{3}{60}, \frac{1}{\sqrt{20}}, \frac{-1}{\sqrt{20}}, \frac{3}{\sqrt{20}}\right)$

$v_2' \left(\frac{-7}{2\sqrt{37}}, \frac{-1}{2\sqrt{37}}, \frac{-7}{2\sqrt{37}}, \frac{7}{2\sqrt{37}}\right)$